

Ryan Cho

Software Engineer | Mobile Authentication & Identity Integration

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PROFESSIONAL SUMMARY

Software engineer specializing in authentication systems, secure API platforms, and integration-ready SDK patterns for regulated financial and enterprise environments. Seven years delivering OAuth 2.0/OIDC flows, centralized login services, token lifecycle management, and mobile security enhancements across distributed remote teams. Proven at CIAM onboarding, headless identity integration, and building production-scale SaaS backends that support native and cross-platform clients.

WORK EXPERIENCE

Software Engineer

May/2022 - May/2026

Aspen Capital

Remote | 10th Floor, New York

- Architected and delivered a **centralized authentication platform** supporting OAuth 2.0/OIDC and JWT validation across a **12-service microservice** ecosystem serving internal and partner applications.
- Built a **headless identity integration layer** with REST and SDK-style endpoints enabling mobile and third-party clients to adopt **single sign-on** without bespoke auth implementations.
- Hardened **session and token lifecycle management** through refresh rotation, short-lived access tokens, and secure cookie policies, reducing auth-related incidents by **40%** over two release cycles.
- Partnered with security and compliance teams on **CIAM onboarding**, mapping identity providers, MFA policies, and audit logging requirements for regulated financial workflows.
- Implemented **mobile security enhancements** including certificate pinning guidance, secure credential storage patterns, and API rate limiting for native client integrations.
- Developed **authentication SDK modules** standardizing login flows, error handling, and telemetry for downstream Android and cross-platform consumers.
- Owned CI/CD pipelines for identity microservices using **Docker, Kubernetes, and GitHub Actions**, cutting deployment time from hours to **15 minutes** with zero-downtime rollouts.
- Mentored **three engineers** on secure API design, threat modeling for auth endpoints, and integration testing strategies for identity-critical paths.
- Collaborated with product and stakeholder groups to define **authentication roadmap** priorities balancing SDK adoption, legacy migration, and regulatory audit readiness.

Technology Analyst, Incoming Technology Analyst, Technology

May/2019 - Apr/2022

Summer Analyst,

BNP Paribas

Remote | Seventh Avenue, New York

- Implemented **enterprise login and SSO integration** connecting internal banking applications to corporate identity directories and federated auth providers.
- Developed **RESTful authentication APIs** consumed by web and mobile channels, enforcing role-based access control across **50+ internal tools**.
- Supported **security enhancement initiatives** including secrets rotation, encryption at rest for credentials, and vulnerability remediation on auth-critical services.
- Built automated test suites for **OAuth flows and permission gates**, improving regression coverage on identity paths by **35%**.
- Assisted cross-team **CIAM migration planning**, documenting integration patterns for application owners onboarding to centralized identity platforms.
- Optimized **token validation middleware** reducing authentication latency by **20%** on high-traffic trading and reporting interfaces.
- Coordinated with distributed teams in **Agile sprints** to deliver authentication fixes and feature parity across multi-region deployments.

Software Engineering Intern

Jun/2018 - May/2019

Chexology

Remote | 17th Floor, New York

- Built **user authentication modules** with email/password and session persistence for a SaaS compliance platform serving **200+ restaurant clients**.
- Integrated **third-party OAuth providers** enabling streamlined merchant login and reducing onboarding friction for new accounts.
- Developed **React-based admin dashboards** with protected routes and role-aware UI components tied to backend permission models.
- Wrote integration tests for **login and registration endpoints**, catching edge-case failures before production releases.
- Collaborated with senior engineers to refactor **API authentication middleware**, improving code reuse across four microservices.

Web Developer

Sep/2017 - Nov/2017

Bayesquare Foundation

Remote | New York

- Delivered **responsive web application** features including user registration, login forms, and profile management for a nonprofit foundation site.
- Implemented **server-side session validation** and HTTPS enforcement improving baseline security posture for public-facing properties.
- Built **REST API integrations** connecting frontend workflows to backend data services with structured error handling.
- Optimized page load performance achieving **30% faster** initial render on authentication-heavy entry flows.
- Partnered with stakeholders to translate **identity and access requirements** into implementable frontend specifications.

EDUCATION

New York University

Bachelor of Arts (B.A.), Computer Science

2016-2020

SKILLS

Frontend: Kotlin, Java, Android SDK, Jetpack Compose, JavaScript, TypeScript, React, HTML5, CSS3

Backend: Node.js, Java, Spring Boot, Python, REST APIs, GraphQL, OAuth 2.0, OIDC, JWT

Databases: PostgreSQL, MySQL, MongoDB, Redis

Cloud & DevOps: AWS, Docker, Kubernetes, GitHub Actions, CI/CD, Terraform

Monitoring & Tools: Datadog, Splunk, SonarQube, Postman, Git, Jira

Workflow & PM: Agile, Scrum, Cross-functional Collaboration, Technical Documentation, SDK Integration